

Code Architecture Design for Dota 2

1. Game Overview

Dota 2 is a multiplayer online battle game. Two teams play against each other. Each team has 5 players. Every player controls one hero with special powers and abilities.

The main goal of the game is to destroy the enemy Ancient building. Players fight enemies, level up, collect gold, and buy items during the match.

Main gameplay features:

Team battles

Hero abilities

Buying items

Multiplayer matches

Towers and map objectives

Real-time combat

2. High-Level Architecture

A big online game needs many systems to work together.

Player System

This system controls:

Player movement

Hero controls

Health and mana

Hero leveling

Combat System

This system handles:

Attacks

Damage

Spells

Fighting mechanics

Inventory System

This system manages:

Items

Gold

Equipment

Item usage

Matchmaking System

This system:

Finds players online

Creates balanced teams

Starts matches

Networking System

This system:

Connects players to the server

Sends game data

Keeps all players synchronized

Game State Management

This system controls:

Match start

Match ending

Winning conditions

Pause and restart systems

UI System

This system shows:

Health bars

Minimap

Inventory

Menus

Scoreboard

3. Code Structure

The game code should be divided into small parts. Every system should have its own job.

Main Classes

HeroController

Controls hero movement and player input.

AbilityComponent

Handles hero spells and cooldowns.

HealthComponent

Stores health, mana, and regeneration.

InventoryComponent

Manages items and item slots.

CombatManager

Handles attacks and damage calculations.

NetworkManager

Sends and receives multiplayer data.

GameManager

Controls game rules and match flow.

UIManager

Updates menus and player interface.

Communication Between Systems

All systems work together.

Examples:

Combat System tells Health System when damage happens.

Inventory System updates the UI when items change.

Network System sends player actions to the server.

Game Manager controls the whole match.

Simple Diagram

GameManager



Player System → Combat System → Health System



Inventory System



UI System



Network System

4. Design Decisions

I selected this architecture because Dota 2 is a very large online game. Many players play together at the same time I've played it myself too.

This design is good because:

It keeps the code organized.

Systems are easy to manage.

New heroes and items can be added easily.

Bugs are easier to fix.

Multiplayer networking works better.

Different developers can work on different systems.

A component-based system is useful because many heroes can reuse the same components like health, abilities, and inventory.

Conclusion

Dota 2 needs many systems to work together smoothly. Using modular systems and component-based design makes the game easier to build and maintain. This architecture helps the game run properly for multiplayer battles and online gameplay.